

1 1.0 Purpose

This work instruction (**Plant Controls FAT Planning, Execution & Handoff CST 7.3-1008 REQ**) defines the process used by the Control Systems Technology (CST) to test, validate the software and hardware for a Distributed Control System (DCS). This work instruction applies to Power & Water Engineering Division (PWED) personnel assigned to support the OTR (Order-to-Remittance) phase of a DCS Project.

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2 2.0 Scope

This work instruction outlines the process used by the CST group to test, validate the software and hardware for a Distributed Control System (DCS).

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3 3.0 Instruction

This section describes the steps performed for the System Software Test, & Validation.

NOTE: The ICS_Checklist.xls file - see Section 6.0 for document hyperlink - is a checklist used by each DCS project to track the completion of a defined set of activities to ensure consistent project execution. This is meant to provide a reminder to mark an item complete in the project's ICS_Checklist.xls file as each item in this Work Instruction is completed.

3.1 3.1 Input Requirements

The following documents and data are inputs required for the completion of this process:

- Network Topology Drawing for FAT
- Performance Monitor Requirements

3.2 3.2 Instructions and References

Table 3.2-1 provides a description of the activities to perform the test, validation, and software packaging of a DCS project, while **Figure 0-1 System Software Test & Validate Map** provides a process map of the activities and illustrates the interface with other groups. Note that the order of the steps is only meant to indicate a general sequence of the activities, but many of the steps can and should be performed in parallel.

Table 3.2-1 System Software Test, & Validate Instructions and References

S t e p #	Description	Additional Instructions and References	ICS Che ck- list
1. 0	Review Preparing for the FAT (FAT Readiness) document.	<u>Preparing for the FAT (FAT Readiness)</u> - see Section 6.0 for document hyperlink. This document provides guidance on preparing for, staging, and conducting the activities for a successful internal Pre-FAT, Customer FAT, and Post-FAT. The document should be used in conjunction with the project's DCS FAT Procedure developed in Step 1.3.	

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S t e p #	Description	Additional Instructions and References	ICS Che ck- list
1. 1	If Performance Monitoring is required, submit requirements to the assigned Performance Monitor engineer.	The Performance Monitoring Requirements need to be submitted to the assigned Performance Monitor engineer prior to Pre-FAT.	
1. 2	Review the DCS part B specification for Integrated FAT with GT, ST Scope	Make sure to verify the Part B spec and with the Project Engineer/Project Manager whether the FAT needs to have GT/ST ordered equipment in the FAT or not. Also, for Software-FAT, Discuss with the OTR System Integration Leader (SIL)/ GT, ST Requisition Engineer for co ordinating the deliverables and schedules to meet the FAT dates	
1. 3	Create the Project DCS FAT Procedure and send to customer for approval.	In addition to core control checks you should also test for network failure latency (power off a switch), external communication devices. <u>DCS FAT Procedure Template</u> - see Section 6.0 for document hyperlink. When considering a Controller only FAT, ensure that any panels that go through IO Express Lane receive the same functional and communication tests as the controllers in FAT. Refer to: <u>Plant Controls IO Express Lane and Hardware Addressing (CST 7.3-1009 REQ-</u> see Section 6.0 for document hyperlink.	

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S t e p #	Description	Additional Instructions and References	ICS Che ck- list
1. 5	Inform HMI Lab/FAT Team Leader if customer equipment is to be used for the FAT.	Inform the HMI Lab/FAT Team Leader in advance that customer supplied equipment is coming for the project. Notify the customer to clearly mark the boxes being shipped with the project name, shop order, and "For FAT Use". Have the Customer send you and the HMI Lab/FAT Team Leader all tracking information. Have the Customer provide you and the HMI Lab/FAT Team Leader all return address information, including a contact at the customer site. Refer to: Shipping Customer Equipment into Salem for FAT - see Section 6.0 for document hyperlink.	
1. 6	Create job directory structure (Master, Site folders) at your desk.	Make sure you enter IP addresses and computer name of destination HMI's (not your own PC name)	

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Step #	Description	Additional Instructions and References	ICS Checklist
1.7	Order product CD and Dongles (HMI and HST) for loading on HMI & Historian computers	NOTE: Technical Leader approval is required for Installation and use of Beta version software (ControlST, eTCSS). eTCSS is required for projects that contain Mark VI devices, i.e. Exciters, Static Starters (Usually for the turbine software) To commence the ordering process, fill out the SupportCentral Workflow Form for requesting Product CDs and USB Keys per the following How-To: <u>CD and Dongle Ordering.doc</u> - see Section 6.0 for document hyperlink. <u>GEIP CCS Product Platform Software Distribution</u>- see Section 6.0 for document hyperlink. Once the workflow has been completed, you can order the product CDs from the site: <u>Controls COE Software Product CD</u> - see Section 6.0 for document hyperlink. For projects where the OSIsoft™ PI Historian is being ordered, select the Turbine Historian Reports Package CD (DS218THR) from the Controls COE Software Product CD site.	
1.8	Provide network topology drawing to HMI Setup / FAT Team Leader so staging area can be planned and setup.	Provide Project 4108, Project Read Me file for the configuration information for the FAT setup team	

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S t e p #	Description	Additional Instructions and References	ICS Che ck- list
1. 9	Work with the HMI Setup / FAT Team Leader to plan and create FAT Layout Diagram	The HMI Setup / FAT Team Leader creates the FAT Layout Diagram for planning of the equipment layout by functionality with input from the Project Lead. HMI Setup / FAT Team will setup the staging area based on the FAT Layout Diagram.	
1. 1 0	Configure HMI computers and copy ToolboxST Application Code and Screens from development environment to FAT computers.	<u>HMI Configuration with ControlST and Cimplicity</u> - see Section 6.0 for document hyperlink.	
1. 1 1	Review/ Follow the Cyber security/ Security*ST Testing Procedure	Review & Follow the Security*ST Maintenance manual GEH-6765.pdf & FAT Procedure document to setup and test the product. - - See section 6.0 for hyperlink	
1. 1 2	Configure backup scripts, time masters, time syncs, global alarm ack, and asset management.	<u>Setting Up Screen Copy Directory and Master Directory.doc</u> - see Section 6.0 for document hyperlink.	

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Step #	Description	Additional Instructions and References	ICS Checklist
1. 1 3	Build system to verify software comm. and controllers running appropriate software	Refer to the ControlST Documentation list installed along with ControlST for Configuration instructions. (For ControlST 5 or above , Refer Mark*Vie Control Initial Startup, GEH-6724A) (For ControlST beore 5 , Refer ControlST Software Suite How-to Guides - GEH-6808) Latest version of above documents can be downloaded from GEIP User Documentation - Salem - see section 6.0 for hyperlink (Navigate from programs->GEControlST-> Documentation) Initial Downloading of a Mark Vle Controller.doc - - see Section 6.0 for document hyperlinks	
1. 2	Pre-FAT: Internal FAT As part of the Pre-FAT, hardware shall be tested to ensure at a minimum that at least 1 I/O point per card is tested, injecting an input and looking at the output for expected values. Hardware checks shall conform to the latest hardware test procedures in Step 1. above.	GEH-6721 Mark Vle Control System Guide - Volume III – Diagnostics and Troubleshooting - - see Section 6.0 for document hyperlink. Additionally, the PACK diagnostic alarms shall be reviewed in accordance with the project specific FAT Procedure. The PPDA PACK diagnostic parameters should be configured to remove nuisance alarms.	

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Step #	Description	Additional Instructions and References	ICS Checklist
1.3	Historian Installation & Configuration (if applicable): Step A: OSIsoft™ PI Software Installation is performed by the HMI Lab / FAT Team. Step B: Configure Historian consisting of: • PI Database • PI ProcessBook Screens and Trend-Charts. • GE Turbine Historian Reports Package (Reports configured) • PI Combo installation on HMI's	GEH-6744 Mark VIe Control Historian System Guide - see Section 6.0 for document hyperlink. Download the Historian Configurator automation tool to create and modify PI ProcessBook: Historian Configurator Tool - see Section 6.0 for document hyperlink.	
1.4	Review CIMPLICITY screens and verify that no screens contain BLACK data. Verify no unconfigured points/objects on CIMPLICITY screens exist. This can be identified by objects as shown below, or points showing a "Red X," which indicate points are not getting data. Document all Red X's that cannot be resolved due to external definitions missing.	If BLACK data occurs on a CIMPLICITY screen due to missing customer deliverables, document the screen and cause for discussions during the FAT. Go to Point Control Panel in CIMPLICITY to list all Red X's on a screen by screen basis. It is important to document all Red X's that cannot be resolved during the FAT, so they can be provided to the site for resolution during installation and commissioning.	

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S t e p #	Description	Additional Instructions and References	ICS Che ck- list
1. 5	Change software as necessary and re-test in FAT environment		
1. 6	Perform customer FAT (get customer signoff)	Any software changes that can be made - without affecting any tests - during the FAT should be implemented. Otherwise, the changes must be implemented after the FAT. Change order process for the project shall be followed to assure changes are properly identified, tracked, and paid for by the end customer. Project engineer and project manager, if assigned, must be involved during this process to approve or reject changes.	
1. 1 9	Fill out the FAT Release Form and obtain customer signature. Have customer complete Customer Survey template	Hardware is only shipped at customer sign off. The FAT Release Form should be returned to the HMI Lab/FAT Team Leader so the equipment can be prepared for shipment (prep-to-ship). The FAT Release Form, Customer survey template for FAT, SWAT can be obtained from the SupportCentral site: <u>GSCM Salem FAT –HMI LAB</u> - see Section 6.0 for document hyperlink.	
1. 2 0	Address post FAT punch list	Before making any changes, receive the Project Engineer's or Project Manager's advice/approval as it may also involve claiming extra engineering hours from customers. Also consider customer specifications, any exceptions taken, and any contractual obligations.	
1. 2 1	Security*ST breakdown & Reshipping procedure	Follow the Security*ST Breakdown/Reshipping procedures before powering down the HMIs	
1. 2 2	Backup and replicate Software (if required)	If time permits perform this step before you run the internal FAT to work out any bugs/issues with search/replace, etc.	

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Step #	Description	Additional Instructions and References	ICS Checklist
1. 2 3	Backup and replicate Screens (if required)	Use the <u>Automationbox</u> Automation Tool to search and replace object name, object variable values (e.g. "CTRLR"). Latest version is available at: <u>Automation Tool Downloads</u> - see Section 6.0 for document hyperlink. Help on use of the tool is also available in the installation folder (default – "C:\Program Files\ReqApps\Trace") once the tool is run first time.	
1. 7	Transmit compliance paperwork to customer.	This will ensure that the compliance paperwork is reviewed by the P&L customer with enough time to transfer custody as planned. Information is transmitted via the agreed upon method defined by the customer. For Ex: Documentum - see Section 6.0 for document hyperlink.	

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S t e p #	Description	Additional Instructions and References	ICS Che ck- list
1. 2 5	Store final software	Archive the Mark*Vie system & zip the screens folder. Store them at: 1) The default archive location on the: HMI Archive Folder - see Section 6.0 for document hyperlink. 2) Project's folder on the: ICS Server Folder - see Section 6.0 for document hyperlink.	
1. 2 6	Inform HMI Lab / FAT Team Leader project is ready to pack and ship.	After FAT is completed, the equipment must be turned over to the HMI Lab / FAT Team for packaging and final shipment. Perform a FAT Release walk through with FAT Tech/ Engineer to verify that all the released FAT Hardware and Software items –including Software packages, Dongles, CD's - are released, returned, and ready for prep to shipment. Verify that there are no missing or incomplete items. If not already provided, send the FAT Release Form, Customer Survey template from Step 1. to HMI Lab / FAT Team Leader.	

Figure 0-1 Plant Controls FAT Planning, Execution & Handoff Process Map

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4 4.0 Roles / Responsibilities

- HMI Lab/FAT Team – Responsible for transfer of equipment from manufacturing to FAT staging area, setup of HMI and network equipment, powering of equipment, setup for test, and tear down, packaging, and shipment of equipment at end of FAT.
- Project Lead - Responsible for the overall fulfillment of customer and regulatory requirements for a given project. Coordinates required activities between the Plant Controls team and GSCM Manufacturing, Finance, Sourcing, and representation at NORM meetings. Responsible for the ongoing project duties .

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5 5.0 Quality Records

The following records produced by the processes covered by the above described work instruction are considered Quality Records and should be stored with the project information. These may not be the only deliverables or documents generated throughout this design process, but are a representative set of descriptive material guaranteeing accurate and consistent quality. Quality Records shall be handled and stored in accordance with P&W QMS 4.2.4.

- FAT Customer Scorecard
- FAT Engineering Scorecard
- FAT Release Form
- I/O Hardware FAT Test Results and acceptance sheets (if early I/O shipment option chosen)
- Project Specific ICS_Checklist.xls
- Signed DCS FAT Procedure and agreement to ship with any fat test results

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6 6.0 References

The following is a list of all references within this document:

- [CD and Dongle Ordering.doc](#)
- [GE IP CCS product platform software distribution](#)(Refer to instructions for ordering can be found here)
- [CD, OSI-PI Ordering link \(Instructions for ordering Dongles is listed\)](#)
- [GSCM SALEM FAT – HMI LAB](#)
- [Controls COE Software Product CD](#)
- [DCS FAT Procedure Template](#)
- [GEH-6765](#)
- [Documentum](#)
- [Initial Downloading of a Mark VIe Controller.doc](#)
- [GE IP User Documentation - Salem](#)
- [PW-003 P&W Quality Dictionary](#)
- [GEH-6721 Mark VIe Control System Guide - Volume III – Diagnostics and Troubleshooting](#)
- [GEH-6744 Mark VIe Control Historian System Guide](#)
- [Historian Configurator Tool](#)
- [HMI Archive Folder](#)
- [HMI Configuration with ControlST and Cimplicity](#)
- [ICS_Checklist.xls](#)
- [ICS Server Folder](#)
- [Plant Controls IO Express Lane and Hardware Addressing \(CST 7.3-1009 REQ\)](#)
- [Preparing for the FAT \(FAT Readiness\)](#)
- [Procedure Improvement Workflow](#)
- [Setting Up Screen Copy Directory and Master Directory.doc](#)
- [Shipping Customer Equipment into Salem for FAT](#)
- [Cyber Security Management System FAT Procedure](#)
- [Security*ST Breakdown/Reshipping procedures](#)

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7 7.0 Definitions and Acronyms

7.1 7.1 Definitions

- Historian – A data archival system, based on OSIsoft™ PI client-server technology. Provides data collection and storage of power distribution and auxiliary process data. (GEH-6744)
- ToolboxST – A Windows-based software package used to configure controllers (GEH-6700). Integrated software package that enables DCS configuration, configuration management, system analysis, automation, remote diagnostics, IP protection, and more.

7.2 7.2 Acronyms

- CD – Compact Disc
- DCS - Distributed Control System
- CST –Control Systems Technology
- PWED - Power & Water Engineering Division
- eTCSS – Electronic Turbine Control System Software
- FAT – Factory Acceptance Test
- PWQMS – Power & Water Quality Management System
- GSCM – Global Supply Chain Management
- HMI - Human-Machine Interface
- HST - Historian
- ICS – Integrated Control System
- IO – Input/Output
- IP – Internet Protocol
- TCW – Turbine Control Workbook (ToolboxST main file)

8.0 Process Management

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